AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

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Presentation Outline

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Problem Statement

Farmers often identify crop diseases only after visible symptoms appear, leading to crop losses, reduced income, and excessive or incorrect pesticide use. Existing AI systems can detect diseases in individual plants, but identifying and tracking possible outbreaks across neighbouring farms remains a challenge. Farmers may also lack timely, location-specific warnings that consider local weather conditions and nearby disease reports. Our team aims to solve this problem through **AGRISYNC – Hyper-Local Crop Disease Outbreak Predictor**, a proposed AI-based system that combines crop disease detection, local weather data, and geographically tagged farmer reports to identify potential outbreak hotspots and alert nearby farmers early, helping them take preventive action and protect their crops.

Problem Understanding

- Farmers, especially small-scale farmers, are most affected by late disease detection and a lack of timely outbreak warnings. Nearby farmers and agricultural officers also face challenges in monitoring disease spread across farms.

- The problem is caused by delayed disease identification, limited access to timely agricultural guidance, changing weather conditions, and a lack of coordination between neighbouring farmers to report and monitor crop disease outbreaks.

- Crop diseases can cause reduced crop yields, financial losses for farmers, and increased pesticide costs. Nearby farmers may face the risk of disease spreading to their fields, while agricultural officers may struggle to monitor outbreaks and provide timely support.

Existing Gaps & Challenges

- Limitation 1: AI disease detection may be less accurate when used on real farm images that differ from the training dataset.
- Limitation 2: Existing systems often focus on detecting diseases in individual plants rather than identifying and tracking potential outbreaks across neighbouring farms.
- The main challenge is to accurately identify and predict potential crop disease outbreaks across neighbouring farms using real-time disease reports and local weather data, so farmers can receive timely warnings and take preventive action.

Our Proposed Solution

- AGRISYNC is an AI-based system that detects crop diseases, combines local weather data and farmer reports to identify potential outbreak hotspots, and sends early warnings to nearby farmers.
- Farmers upload crop images and location details to AGRISYNC. The AI detects possible diseases, while local weather data and nearby farmer reports help identify potential outbreak hotspots. The system then sends early warnings and preventive guidance to affected farmers.
- AGRISYNC helps farmers detect crop diseases early, receive timely outbreak alerts, reduce crop losses and unnecessary pesticide use, and protect their income through preventive action.

Technical Approach

image.png

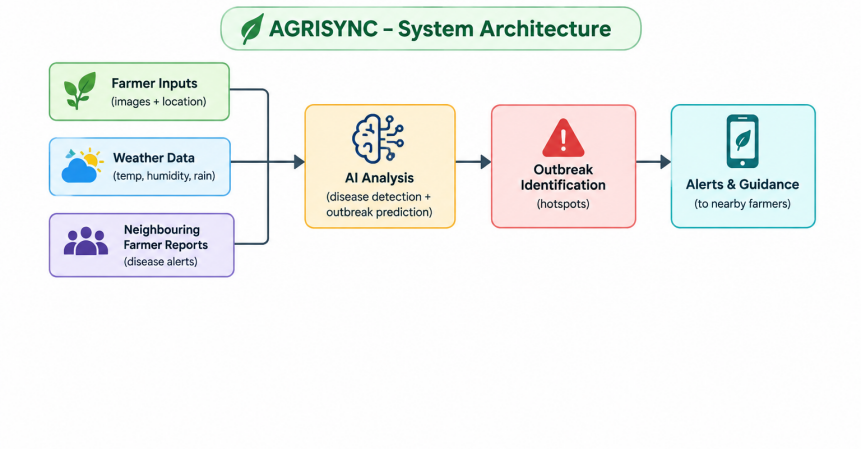


Figure: System Architecture of AGRISYNC

Innovation & Uniqueness

- AGRISYNC combines AI-based disease detection, local weather data, and neighbouring farmers' reports to identify potential outbreak hotspots and send early warnings to nearby farmers, rather than focusing only on individual plants.
- This approach is suited to the problem because it combines disease detection, weather conditions, and nearby farmer reports to identify potential outbreaks early, helping farmers take preventive action, reduce crop losses, and avoid unnecessary pesticide use.
- AGRISYNC's key differentiator is its hyper-local outbreak monitoring, which combines AI disease detection, weather data, and nearby farmer reports to alert neighbouring farms about potential disease outbreaks early.

Feasibility & Scalability

- **Feasibility:** Resources: AI disease detection model, crop image datasets, weather data, farmer reports, a mobile app, database, and internet access. Steps: Collect data, train the AI model, integrate weather and location data, develop the app, test outbreak alerts, and deploy the system for farmers.]
- **Scalability:** AGRISYNC can expand to more villages and regions, support multiple crops and local languages, integrate with agricultural experts, and partner with government agencies and farming organizations to provide wider, more accurate outbreak alerts.
- **Risks:** Constraints: Limited internet access, inaccurate AI predictions, insufficient farmer reports, and difficulty collecting reliable data. Mitigations: Provide offline-friendly features, train AI with diverse crop images, encourage farmer participation, and validate predictions with agricultural experts.

Cost for Product (Optional)

- Development or hardware cost: 10,000–30,000 for a basic prototype using existing AI models and free cloud services. Hardware costs can be minimized by using students' existing laptops and smartphones.
- Estimated Hosting and Maintenance Cost: 500–2,000 per month for basic cloud hosting, database storage, bug fixes, and regular AI model updates. Costs may increase as the number of users and data grows.
- Estimated Total Cost: 13,000–54,000 for initial development and the first year of basic hosting and maintenance.

Use Cases (compulsorily 3 use cases must be there)

- Use case 1: Farmer uploads a crop image to detect possible diseases and receive preventive guidance.
- Use case 2: Nearby farmer receives an early warning about a potential disease outbreak based on local weather and farmer reports.
- Use case 3: Agricultural officer monitors reported disease hotspots and helps farmers take appropriate preventive measures.
  Note: Additional slides may be included only for this section.

Impacts and Benefits

- AGRISYNC helps farmers detect crop diseases early, receive location-specific outbreak alerts, reduce crop losses and unnecessary pesticide use, and make better farming decisions. It also helps agricultural officers monitor potential outbreaks and provide timely support.

- Social Impact: Helps farmers access timely disease alerts and improve awareness.
  Economic Impact: Reduces crop losses and unnecessary pesticide expenses, helping protect farmers' income.
  Environmental Impact: Encourages targeted pesticide use and supports healthier soil and ecosystems.

- AGRISYNC aims to improve early crop disease detection and deliver timely alerts to nearby farmers. It seeks to reduce preventable crop losses and unnecessary pesticide use. Success will be measured through detection accuracy, alert response time, and changes in crop losses.

Sustainable Development Goals (SDGs)

SDG NUMBER	Target Number	Goal Name	Connection
SDG [2]	Target [number]	[Zero Hunger]	[AGRISYNC helps farmers detect crop diseases early and reduce crop losses. It promotes sustainable farming and supports food security.]
SDG [12]	Target [number]	[Responsible Consumption and Production]	[AGRISYNC reduces unnecessary pesticide use and promotes sustainable farming through early disease detection.]

Individual Member Contribution

Member	Contribution or responsibility
[Member 1 Sujay Saravanan]	[Researched the problem statement, existing solutions, and limitations.]
[Member 2 T Sai Thanush]	[Designed the AGRISYNC concept, system architecture, and workflow.]
[Member 3 Poornesh]	[Identified the required technologies, development costs, and implementation steps.]
[Member 4 Umesh Dhilly]	[Studied the expected benefits, SDG connections, and future expansion possibilities.]

Yukti ID

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